

Composition error propagation for the noisy folded limit cycle

the composed chart map concentrates, θ -uniformly: errors add and the fold renews

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Abstract

We prove the propagation step of Path A: the composed noisy transition map $\Pi = \Pi_3 \circ \kappa_{23} \circ \Pi_2 \circ \kappa_{12} \circ \Pi_1$ across the blow-up charts K_1 (entry) $\rightarrow K_2$ (inner/fold) $\rightarrow K_3$ (exit) concentrates around the deterministic JKK passage with errors that *add* rather than compound, uniformly in the angle $\theta \in S^1$. The mechanism is a telescoping (discrete Gronwall) bound for the transverse deviation together with the *renewal* supplied by the fold: the entry-to-peel-off map is an exponential contraction, so any entry deviation is forgotten and cannot accumulate across K_2 . Consequently the exit law is the inner exit measure (Tracy–Widom, established separately) transported by the bounded exit chart, plus sub-dominant Gaussian corrections, and the critical-noise law $\sigma_*^A = 2\sqrt{\pi} \sqrt{\varepsilon_2} \sqrt{ac/b}$ survives the composition. The argument reduces the full composition to per-chart inputs that are either standard Berglund–Gentz concentration (K_1, K_3) or already proved (K_2 inner exit measure, forgetting, θ -uniformity).

1 Framework

After blow-up the passage is a finite chain of slow–fast SDE segments. We track the *transverse coordinate* x (the deviation of the noisy trajectory from the deterministic canard γ) across a sequence of sections $\Sigma_0 \rightarrow \Sigma_1 \rightarrow \Sigma_2 \rightarrow \Sigma_3$, with Σ_0 the entry to K_1 , $\Sigma_1 = \Sigma_{12}$ the K_1/K_2 overlap, $\Sigma_2 = \Sigma_{23}$ the K_2/K_3 overlap, and Σ_3 the exit. The chart-overlap maps $\kappa_{i,i+1}$ (smooth blow-down Jacobians) are absorbed into the transition maps. Writing x_i for the state on Σ_i and $\bar{x}_i$ for the deterministic state, each stage is a random map

$$x_i = f_i(x_{i-1}) + \zeta_i, \quad \bar{x}_i = f_i(\bar{x}_{i-1}), \quad (1)$$

where f_i is the deterministic transition (JKK) and ζ_i the noise increment of the i th segment. By the strong Markov property the ζ_i are generated by disjoint portions of the driving Brownian motion, hence conditionally independent given the x_{i-1} . Set $\delta_i := x_i - \bar{x}_i$ (the accumulated transverse error).

We will use two structural inputs that are *proved* elsewhere.

Input 1 (θ -uniform deterministic bounds). Each f_i is Lipschitz with constant L_i , and the L_i together with the contraction and quasipotential constants are bounded above and below by θ -independent constants depending only on $\min / \max$ of a, b, c on S^1 . (*Proved: θ -uniformity lemma; the deterministic maps and their derivatives are JKK Theorem 3.2.*)

Input 2 (fold renewal / forgetting). The entry-to-peel-off map (the composition K_1 then the attracting phase of K_2 up to the fold) is an exponential contraction: its Lipschitz constant is $\Lambda_c \leq e^{-\frac{4}{3}Y_*^{3/2}/1}$ (contraction rate $2\sqrt{Y}$ integrated to the entry level Y_*), so $\Lambda_c \rightarrow 0$. Equivalently $L_2 \leq \Lambda_c$ in (1). (*Proved: the canard is attracting; this is the “forgetting” Lemma R1 used in the shooting characterisation, and is confirmed numerically—the peel-off law is invariant to entry offset/noise within the basin.*)

2 The propagation lemma

Lemma 1 (errors add). *For the chain (1),*

$$|\delta_m| \leq \left(\prod_{i=1}^m L_i \right) |\delta_0| + \sum_{j=1}^m \left(\prod_{i=j+1}^m L_i \right) |\zeta_j|.$$

Moreover, if $\mathbb{P}(|\zeta_j| > h_j \mid x_{j-1}) \leq p_j(h_j)$ for the relevant entry range, then for any tube radii h_j ,

$$\mathbb{P}\left(|\delta_m| > \left(\prod_i L_i\right) |\delta_0| + \sum_j \left(\prod_{i>j} L_i\right) h_j\right) \leq \sum_{j=1}^m p_j(h_j).$$

All constants are θ -uniform by Input 1.

Proof. From (1), $\delta_i = x_i - f_i(\bar{x}_{i-1}) = \zeta_i + (f_i(x_{i-1}) - f_i(\bar{x}_{i-1}))$, so by the Lipschitz bound $|\delta_i| \leq |\zeta_i| + L_i |\delta_{i-1}|$. Iterating this scalar recurrence (discrete Gronwall) gives the deterministic estimate. For the probabilistic bound, the event $\{|\delta_m| > \dots\}$ is contained in $\bigcup_j \{|\zeta_j| > h_j\}$ (if every $|\zeta_j| \leq h_j$ the deterministic estimate gives the displayed bound); the union bound and the conditional tails p_j , together with the tower property over the Markov chain, yield the claim. Uniformity in θ is inherited from the θ -uniform L_i and p_j (Input 1). $\square$

Remark (why errors do not compound). The only way the chain could blow up is through $\prod_i L_i \gg 1$. Two facts forbid it: the products are θ -uniformly bounded (Input 1; JKK’s composed map is a diffeomorphism with controlled derivatives), and—decisively—the fold factor obeys $L_2 \leq \Lambda_c \rightarrow 0$ (Input 2). Hence every product that contains the fold stage is exponentially small. This is the *renewal*: the fold resets the transverse memory, so deviations entering K_2 are annihilated, not amplified.

3 Per-chart inputs

Input 3 (K_1 , attracting Fenichel tube; Berglund–Gentz). On K_1 the deviation from the attracting slow manifold is an Ornstein–Uhlenbeck-type process with θ -uniform contraction rate κ_1 ; for $h > 0$, $\mathbb{P}(\sup |\zeta_1| > h) \leq C_1 (t_1/\eta^2) e^{-\kappa_1 h^2/2\eta^2}$, and f_1 is a contraction onto γ . [1, Ch. 5].

Input 4 (K_2 , inner exit measure). The peel-off level on Σ_2 has the inner exit measure μ_η : in the inner scaling $(Y, R) = (\eta^{4/3}y, \eta^{2/3}r)$ it is the law of $-\Lambda_0(\beta) \stackrel{d}{=} \text{TW}_\beta$, $\beta = 4/\eta^2$, independent of the entry (Input 2); the early-escape probability is $1 - F_\beta(0)$. Its tail is the TW_β tail, so $\mathbb{P}(|\zeta_2| > h) \leq p_2(h)$ with p_2 the (Painlevé II) Tracy–Widom tail. (*Proved: reduction theorem and shooting characterisation.*)

Input 5 (K_3 , exit tube; Berglund–Gentz). On K_3 the trajectory, having left the canard, follows the deterministic exit solution with a θ -uniform Gaussian tube: $\mathbb{P}(\sup |\zeta_3| > h) \leq C_3 (t_3/\eta^2) e^{-\kappa_3 h^2/2\eta^2}$, and the exit map f_3 has θ -uniform Lipschitz constant $L_3 \leq \bar{L}$ (the bounded blow-down Jacobian of κ_{23} and the K_3 flow). [1, Ch. 5]; [2].

4 The composition theorem

Theorem 1 (composition error propagation). *Under Inputs 1–5, the composed noisy transition map $\Pi = \Pi_3 \circ \kappa_{23} \circ \Pi_2 \circ \kappa_{12} \circ \Pi_1$ satisfies, with $\delta_0 = 0$:*

- (a) (concentration) *for tube radii h_1, h_2, h_3 ,*

$$\mathbb{P}(|\delta_3| > \bar{L} h_2 + h_3 + \Lambda_c h_1) \leq p_1(h_1) + p_2(h_2) + p_3(h_3),$$

θ -uniformly; the entry contribution $\Lambda_c h_1$ is exponentially small;

- (b) (exit law) *as $\eta \rightarrow 0$, the exit transverse coordinate converges in law to $f_3(\mu_\eta)$ —the inner exit measure transported by the exit chart— uniformly in θ ; the sub-dominant correction is the $O(\eta)$ Gaussian ζ_3 ;*
- (c) (critical noise) *the escape probability of the whole passage equals the inner escape probability $1 - F_\beta(0)$ up to the sub-dominant corrections of (a); hence escape becomes likely at $\sigma_*^A = 2\sqrt{\pi} \sqrt{\varepsilon_2} \sqrt{ac/b}$, uniformly in θ .*

Proof. Apply Lemma 1 with $m = 3$ and the per-chart tails of Inputs 3–5. The Lipschitz products are $\prod_{i=1}^3 L_i = L_3 L_2 L_1 \leq \bar{L} \Lambda_c L_1$ and, in the sum, $\prod_{i>1} L_i = L_3 L_2 \leq \bar{L} \Lambda_c$ (multiplies $|\zeta_1|$), $\prod_{i>2} L_i = L_3 \leq \bar{L}$ (multiplies $|\zeta_2|$), and the empty product 1 (multiplies $|\zeta_3|$). With $\delta_0 = 0$ this gives $|\delta_3| \leq \bar{L} \Lambda_c |\zeta_1| + \bar{L} |\zeta_2| + |\zeta_3|$, and the union bound of Lemma 1 yields (a) (writing $\Lambda_c h_1$ for the exponentially suppressed entry term).

For (b): take $h_1 = O(\eta)$, $h_3 = O(\eta)$ (the Berglund–Gentz tube radii at concentration level), so $\bar{L} \Lambda_c h_1$ and h_3 are $O(\eta)$ or smaller, while ζ_2 carries the inner scale $O(\eta^{2/3})$ (Input 4). Thus $\delta_3 = \bar{L} \zeta_2 + O(\eta)$, i.e. $x_3 = f_3(\bar{x}_2 + \zeta_2) + O(\eta) = f_3(\mu_\eta) + O(\eta)$ in law, uniformly in θ (the $O(\eta)$ is the union of the θ -uniform Gaussian tubes). Because $L_2 \leq \Lambda_c \rightarrow 0$, the entry law x_0 (and its fluctuation ζ_1) drops out: the fold renews the chain (Input 2), so the limit is independent of the entry data.

For (c): the passage escapes iff the canard peels off before the fold, an event of the inner chart with probability $1 - F_\beta(0)$ (Input 4); by (a) the surrounding charts perturb this only by the union-bound corrections $p_1 + p_3 = O(e^{-c'/\eta^2})$, which are sub-dominant to $1 - F_\beta(0)$. The leading-order threshold is unchanged: escape becomes order-one when the integrated inner hazard $H = \eta^2/4\pi$ is order one, i.e. at $\eta_* = 2\sqrt{\pi}$, giving σ_*^A . All bounds are θ -uniform. $\square$

5 Scope

What is proved here. The *propagation* itself: that the composed map’s transverse error is the θ -uniformly weighted sum of per-chart fluctuations (Lemma 1), that the fold renewal annihilates the entry contribution (so errors do not compound across the non-hyperbolic chart), and that the exit law and the critical-noise threshold are therefore the inner ones transported by the bounded exit chart. This is exactly the “error accumulation, not just per-factor” step that was the last structural gap in Part 2.

What is imported. The per-chart concentration on the *normally hyperbolic* charts K_1, K_3 is standard Berglund–Gentz sample-path concentration [1, 2]; the inner (K_2) chart’s exit measure, the forgetting, and the θ -uniformity are proved in the companion notes. The deterministic composed map and its derivative bounds are JKK Theorem 3.2.

What remains. (i) The blow-*down* from the blow-up charts to the physical $(\varepsilon_1, \varepsilon_2)$ variables (the final desingularisation bookkeeping, turning the θ -uniform chart estimates into a statement for the original system); (ii) the Channel-B (phase) inner with its bifurcation-specific exponent. Neither affects the propagation proved here.

References

- [1] N. Berglund, B. Gentz, *Noise-Induced Phenomena in Slow–Fast Dynamical Systems: A Sample-Paths Approach*, Springer (2006).
- [2] N. Berglund, B. Gentz, C. Kuehn, *From random Poincaré maps to stochastic mixed-mode-oscillation patterns*, J. Dynam. Differential Equations **27** (2015) 83–136; *Hunting French ducks in a noisy environment*, J. Differential Equations **252** (2012) 4786–4841.
- [3] S. Jelbart, C. Kuehn, N. Kuntz, arXiv:2208.01361 (2024). — deterministic backbone, Theorem 3.2, charts K_1 – K_3 .
- [4] Companion notes: `ShootingCharacterisation_proof` (inner exit measure, forgetting), `InstantonPrinciple_proof` (rate), and `PathA_endgame_writeup` (§ θ -uniformity).